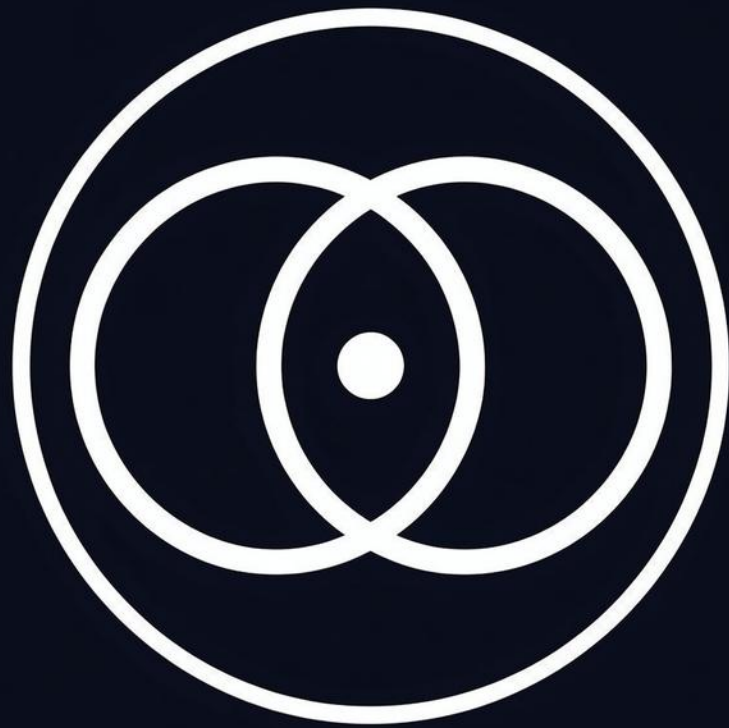




P5

Conformal Prediction Credibility and Parametric Uncertainty Fans for a Small Kinematic World Model

小型世界模型的共形预测可信度与参数不确定性

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Conformal Prediction Credibility and Parametric Uncertainty Fans for a Small Kinematic World Model

Chinese Abstract

小型可微动力学世界模型在边缘端做长程预测时，最容易被高估的是"预测有多可信"。本文报告一个约 96.8 万参数的残差循环世界模型（UDOS v7.0.3，单 CPU、固定 seed），在严格轨迹级四分（训练/验证/测试/校准，独立 seed、无相邻窗口泄漏）下，将**校准概率与熵置信度严格解耦**：按显著性水平 α (0.2/0.1/0.05) 单独取分位数的 split conformal 区间，在 held-out 测试集 80/90/95 三水平上，oracle 边际覆盖为 0.784/0.898/0.958，blind 为 0.776/0.900/0.956，全部落入 ± 0.05 容差；盲路径整体 MSE 0.0104（v7.0.1 基线 0.0459）。参数不确定性以蒙特卡洛自助扇形给出 p10/p50/p90，经两步膨胀后经验覆盖 0.859（名义 0.80，偏保守）。21 项门禁全过。本文不主张共形方法创新，贡献在于：给出一个**可复现、证据分级、负结果**

不遮掩的小型动力学世界模型可信度工程基准，并明确区分"偶然不确定性（共形区间）"与"参数不确定性（蒙特卡洛扇形）"两类带宽。

Keywords (Chinese): conformal prediction; world models; uncertainty quantification; time-series forecasting; reproducibility; edge AI

Abstract

Small differentiable kinematic world models deployed on edge devices most often overclaim how credible their long-horizon predictions are. This paper reports a residual recurrent world model of about 0.97M parameters (UDOS v7.0.3, single CPU, fixed seeds), under a strict trajectory-level four-way split (train/val/test/calibration with independent seeds and no adjacent-window leakage), in which **calibrated probability is strictly decoupled from entropy confidence**. Split-conformal intervals whose nonconformity score is quantile-taken per α (0.2/0.1/0.05) achieve, on held-out test, oracle marginal coverage of 0.784/0.898/0.958 and blind coverage of 0.776/0.900/0.956 at the 80/90/95 nominal levels—all within the ± 0.05 tolerance. Blind overall MSE is 0.0104 versus a v7.0.1 baseline of 0.0459. Parametric uncertainty is exposed as a Monte-Carlo bootstrap fan (p10/p50/p90) that, after two-stage inflation, reaches empirical coverage 0.859 at nominal 0.80 (conservative). All 21 gates pass. We do not claim a conformal-method novelty; the contribution is a reproducible, evidence-graded, no-hidden-failure engineering benchmark for small kinematic world models, and an explicit separation between aleatoric interval width and parametric fan width.

Keywords: conformal prediction; world models; uncertainty quantification; time-series forecasting; reproducibility; edge AI

Structured Abstract (background/problem → method → evidence/results → contribution)

- **Background and problem:** small differentiable kinematic world models most often overclaim how credible their long-horizon predictions are. The entropy, energy, or logit temperature output by a neural forward pass is, before calibration, only a relative-scale ranking signal, not to be taken as "I am 90% confident" to drive downstream safety decisions; on edge the model must also be small, fast, interpretable, not trading robustness for parameter stacking and GPU ensembles.
- **Method:** on a residual recurrent world model of **967,796** parameters (UDOS v7.0.3, single CPU, fixed seeds), using a strict trajectory-level independent-seed four-way split (train/val/test/calib, no adjacent-window leakage), **calibrated probability strictly decoupled from entropy confidence:** split-conformal intervals quantile-taken per significance level α (0.2/0.1/0.05) describe aleatoric uncertainty, while a Monte-Carlo fan of empirical bootstrap ($M=64$) over calibration-set parameter residuals plus two-stage inflation describes parametric uncertainty, the two widths delivered to separate objects and accepted under separate tolerances.
- **Evidence and results:** the six held-out test groups (oracle/blind $\times$ 80/90/95) marginal coverage fall in **0.7758–0.9577**, all within ± 0.05 ; blind overall MSE **0.0104** (v7.0.1 baseline 0.0459); the parameter fan after two-stage inflation empirical coverage **0.8592** (nominal 0.80, conservative); all 21 automated gates pass. Negative results not hidden: spring blind MSE 0.0186 highest of the four, blind 80% coverage 0.7758 near the tolerance lower edge.
- **Contribution:** no conformal-method novelty claimed; the contribution is—(i) delivering and accepting aleatoric uncertainty (conformal intervals) and parametric uncertainty (Monte-Carlo fans)

as two non-impersonating objects; (ii) a reproducible, evidence-graded (verified/cpu-proto/unverified three levels), no-hidden-failure engineering benchmark for small kinematic world models. **All numbers are CPU fixed-seed single-run pilot point estimates**, no ≥ 30 -seed replication, no p-values/CIs reported; external industry numbers marked unverified.

1 Introduction

Taking a "prediction interval" for a "probability" is one of the most common failure modes of learned kinematic systems at the engineering stage. The entropy, energy, or logit temperature output by a neural forward pass is, before calibration, only a relative-scale ranking signal, not to be taken as "I am 90% confident" to drive downstream safety decisions. When a world model is used in robotics, simulation, or control loops, this misreading of uncertainty as probability translates directly into overconfident planning—while on edge the model must be small, fast, interpretable, not trading robustness for parameter stacking and large GPU ensembles.

This paper faces exactly such a small-model scenario: the core prediction kernel `WorldModelCore` in UDOS Reasoning Engine v7.5.0 has only **967,796** trainable parameters (hidden dim 256, 2 GRU layers, scene dim 32), trained and inferred on two CPU threads. What we care about is not "how low it pushes MSE," but: **when the scene's hidden physical parameters are unknown (blind path), can the prediction intervals it gives reach nominal coverage on truly held-out data? Does interval width vary monotonically with α and grow reasonably with the rollout step? Can parametric uncertainty be characterized separately rather than mixed with aleatoric uncertainty in one entropy value?**

Behind these three questions is a plainer engineering stance: **credibility is not an appendage of point-prediction accuracy, but a first-class indicator requiring separate acceptance**. A model with very low MSE but erratic coverage and a model with slightly higher MSE but reliable intervals have entirely different value in a control loop—the latter can be safely "used at a discount" downstream, the former fails suddenly on out-of-distribution samples. All engineering design here (independent calibration set, quantile per α , separate tolerances for intervals and fans, 21 automated gates) serves this stance.

The evidence grade of external motivation must be stated first. Recent industry progress on "general world models," "real-time video prediction," "embodied foundation models" is fast, but the related numbers (e.g. larger video-prediction models' parameters, frame rates, benchmark scores) are mostly vendor or report caliber, **not independently verified in this study** (below, wherever such external numbers appear, marked unverified, motivation only, not cross-proven with this repo's measured numbers). We deliberately narrow the scope to a **synthetic kinematic world with known ground truth**: uniform, uniform acceleration, simple harmonic motion, 1D elastic collision four classes, 3D directions, 1D collision, state a 6-dim vector `[px,py,pz,vx,vy,vz]`. Known ground truth is both a soft spot (external validity limited to synthetic data, see Chapter 8) and the premise for computing statistics like "coverage," "identifiability" accurately and reporting negative results honestly.

The specific contributions of this paper are as follows:

1. **A method-engineering-decoupled credibility pipeline**: centered on the real source modules `ConformalCalibrator` in `udos7/uncertainty.py` (finite-sample-corrected split conformal with quantile taken separately per α) and `MonteCarloParamFan` (calibration-set parameter residual empirical bootstrap + two-stage inflation), splitting aleatoric uncertainty (prediction intervals) and parametric uncertainty (parameter fans) into two independent, non-impersonating objects.

2. **Strict data splits and gates:** trajectory-level independent-seed four-way split (train/val/test/calib), the calibration set not participating in gradients, not mixed with val/test; any of the 21 automated gates with coverage out of bounds judges FAIL.
3. **Honest benchmark numbers:** blind MSE 0.0104 (v7.0.1 0.0459); six coverage groups at three levels all pass; parameter fan empirical coverage 0.859 (conservative, discussed honestly); and reporting known short boards—spring blind MSE 0.0186 highest, blind 80% coverage 0.776 near the tolerance lower edge.

Boundary statement: all numbers here come from a CPU fixed-seed single verification run (`reports7/v7_verification.json`, evidence grade verified). This is a **pilot point estimate**; except where noted, no ≥ 30 -seed replication, hence no p-values, confidence intervals, or significance conclusions, related positions retaining `[RESULT NEEDED]`.

2 Related Work

Conformal prediction provides finite-sample, distribution-free marginal coverage guarantees for black-box predictions. Its core mechanism: sorting nonconformity scores on the calibration set, taking the quantile at nominal level $1-\alpha$ as the width, hence under the assumption that "calibration and test samples are exchangeable," without any parametric assumption on the data distribution, giving about $1-\alpha$ finite-sample marginal coverage (Angelopoulos & Bates, 2021, arXiv:2107.07511). The boundary of this guarantee is likewise stated by the survey: marginal coverage is not conditional coverage holding for every conditional subgroup, and non-exchangeable, non-stationary data need special correction [CITATION NEEDED: split conformal prediction original source Vovk 2005]. Applying it to time series and autoregressive rolling requires handling two non-trivial problems: one, autoregressive error accumulation in multi-step rolling; two, coverage decay under distribution drift [CITATION NEEDED: conformal time series; conformal prediction under distribution shift; envelope conformal]. Uncertainty quantification for world models / differentiable simulators (neural ODE, latent dynamics) usually distinguishes aleatoric uncertainty (data/process noise) and epistemic uncertainty (model parameters/structure unknown) [CITATION NEEDED: epistemic vs aleatoric uncertainty; uncertainty in world models/neural forecasting].

Three lines are closest to this paper. First, **class-conditional / weighted conformal** handling covariate drift [CITATION NEEDED: class-conditional conformal; weighted conformal for covariate shift]; our oracle path is equivalent to taking the class as an explicit condition, the blind path not using the class—this "knowing/not knowing the scene class" contrast exactly quantifies conditioning's impact on coverage and width. Second, **Monte-Carlo dropout / deep ensembles** approximating epistemic uncertainty [CITATION NEEDED: Monte Carlo dropout Gal Ghahramani; deep ensembles Lakshminarayanan]; we do not use dropout, but **empirical bootstrap over calibration-set parameter residuals**, because our hidden parameters have only 4 slots (initial velocity magnitude, acceleration magnitude, spring angular frequency, collision-pair velocity), bootstrap sampling explicitly preserving inter-slot correlation, and not needing multiple trained models. Third, **world model credibility benchmarks** [CITATION NEEDED: world model benchmark; calibrated uncertainty for learned dynamics].

On the time-series conformal branch, non-exchangeability from autoregressive rolling is the core difficulty: step h 's prediction depends on the previous $h-1$ steps' prediction errors, errors accumulating along the trajectory, hence "per-step coverage" usually decays with h , which is why we report both marginal and per-step widths. Academia already has envelope conformal, conformalized quantile regression, adaptive (ACI) and other drift-handling methods [CITATION NEEDED: adaptive conformal prediction; conformalized quantile regression Romano], this paper does not yet introduce an

adaptive mechanism, but takes "decay under drift" as a known limitation and future work, to keep the current protocol simple and reproducible.

Our positioning is **method application + engineering benchmark**, no conformal-method novelty claimed. Novelty comes mainly from three engineering honesties: (i) strict trajectory-level four-way split and independent calibration seed; (ii) interval coverage and parameter-fan coverage accepted separately, tolerances set separately; (iii) opening a complete pipeline of 0.97M parameters, single-CPU reproducible, together with source, checkpoint, report JSON.

A set of concepts needing special distinction: **conformal intervals characterize the aleatoric uncertainty of "where the true value of a given point prediction falls"** (regardless of parameter right/wrong, width set by residual size), while **parameter fans characterize the epistemic uncertainty of "how much inaccurate hidden-parameter estimation swings the point prediction."** The two cannot substitute: a model can have accurate point predictions (small aleatoric residual, narrow intervals) but biased parameter estimates (wide epistemic fan); and vice versa. This pipeline delivers and accepts the two separately, precisely to avoid the common engineering practice of "using one entropy value to answer both questions."

3 Problem Definition and Assumptions

3.1 Notation and Problem Setup

State vector $\mathbf{s}_t \in \mathbb{R}^6$, convention $[p_x, p_y, p_z, v_x, v_y, v_z]$ (`STATE_DIM=6` in `contracts.py`). Given observation window $X = (s_{t-W+1}, \dots, s_t)$ (window $W=6$), the world model outputs an H -step rollout $\hat{Y} = (\hat{s}_{t+1}, \dots, \hat{s}_{t+H})$, $H=4$. Scene hidden parameters $\mathbf{P} \in \mathbb{R}^4$, slot order fixed as $(v_0, \text{accel}_a, \text{spring}_\omega, \text{other}_{v2})$ (`SCENE_PARAM_NAMES` in `contracts.py`).

- **Oracle path:** forward explicitly injecting ground truth $\mathbf{P}$ (`explicit=P`).
- **Blind path:** $\mathbf{P}$ not provided, `SceneEstimator` estimating $\hat{\mathbf{P}}$ from the window and falling back (`scene.py`).

For a test sample, point prediction $\hat{Y}$, conformal intervals giving $[L, U]$ (per-step width broadcast to 6 state dims). **Empirical coverage** defined as $\text{cov} = \frac{1}{NH} \sum_{n,h,d} \mathbb{1}\{L_{n,h,d} \leq Y_{n,h,d} \leq U_{n,h,d}\}$, $\text{tag}[1]$ with per-step and per-mode calibers. **Marginal width** q_α is the finite-sample quantile of nonconformity scores at that α .

"Per-mode" meaning: oracle and blind each fit their own width from their own calibration-set residuals, not sharing one width. This is because the two paths' nonconformity distributions differ—oracle knows the class, residuals more concentrated; blind estimates parameters, residuals more spread. Sharing a width would make oracle too wide or blind too narrow. This implementation builds one `ConformalCalibrator` instance per path, separately `fit`, which is also why Table 2 has six rows rather than three.

3.2 Falsifiable Hypotheses

- **H1 (marginal coverage passes):** within tolerance ± 0.05 , at the three nominal levels 80/90/95 both oracle and blind marginal empirical coverage fall in the tolerance band. Falsifiable: any (mode, α) with $|\text{cov} - \text{nominal}| > 0.05$ rejects.

- **H2 (width monotonic with α , blind wider than oracle)**: width grows as nominal coverage rises (α falls); at the same α blind width not narrower than oracle. Falsifiable: $\exists \alpha$ making width non-monotonic, or blind width < oracle width.
- **H3 (parameter fan reaches nominal)**: the blind parameter fan after two-stage inflation, the [p10,p90] central band's empirical coverage on test within ± 0.10 of nominal 0.80. Falsifiable: $|\text{cov}_{\text{fan}} - 0.80| > 0.10$.
- **H4 (entropy confidence is not probability)**: the deterministic kernel outputs no probability; any taking entropy for coverage is uncalibrated and worsens with distribution drift. This paper executes "no entropy probability, only conformal/fan delivery," coverage decay under drift as a future experiment (see Chapter 9).

4 Method and System Design

4.1 Prediction Kernel: Residual Recurrent World Model

`WorldModelCore` in `udos7/model.py` is a single inference graph, no second engine, no LoRA. Its structure: observations frame by frame encoded by `obs_encoder` (Linear-LayerNorm-GELU two segments, input dim `STATE_DIM=6` $\rightarrow$ hidden 256), each frame injecting unified scene context `ctx` (produced by `SceneChannel`, dim $32 \rightarrow 256$ via `ctx_proj`), then fed into a 2-layer GRU temporal kernel (hidden=256, `num_layers=2`, `batch_first=True`), residual decoding (decoder: Linear-GELU-Linear, $256 \rightarrow 256 \rightarrow 6$) $s_{t+1} = s_t + \Delta(h_t)$. The residual form makes "uniform continuation" a zero-learning-cost baseline, the model only needing to learn acceleration/oscillation/collision deviations; its advantage over the old CTM tick stacking shows in `improvement_vs_v701` as blind overall MSE falling from 0.0459 to 0.0104. The decoder's last layer zero-initialized, making the initial model strictly predict "state unchanged," training starting from a stable point.

v7.0.3 also adds an independent small gate `kin_gate` eating only deterministic kinematic features (Linear(`KIN_DIM=11,32`)-GELU-Linear($32,6$), last layer zero-init), softly mixing analytic constant-acceleration integration into the prediction by `ca_conf` (constant-acceleration consistency confidence, $\in [0,1]$): $s_{t+1} = (1-g)s_t + g \cdot \text{MLP}(\text{kin})/2$, where `analytic` uses the observable acceleration vector $a \in \mathbb{R}^3$ fixed from the initial window for $v_{t+1} = v_t + a \cdot dt$, $p_{t+1} = p_t + v_t \cdot dt + \frac{1}{2} a \cdot dt^2$. This gate does not read the shared GRU hidden state—the design intent being to avoid the gate, after saturating on uniform/accel, changing the shared representation gradients and dragging spring's learning path. `ca_conf` itself given by `kinematics.py`: per-window R^2 of velocity linearly fitted against time, then multiplied by $(1 - \text{spring_valid})(1 - \text{jump_valid})$, making uniform/accel ≈ 1 , spring/collision ≈ 0 . Measured gate means (`improvement_vs_v703.gate_mean_by_kind`): uniform 0.634, accel 0.662, spring 0.0, collision 0.154, consistent with design intent—the gate fully closed on spring, cautious on collision (no jump yet observed within the window but collision within horizon).

4.2 Conformal Calibrator: Quantile Taken Separately per α

`ConformalCalibrator` in `udos7/uncertainty.py` fixes the old version's defect where "nominal 80/90/95 empirical coverage all the same constant, α ignored." This old bug is worth recording because it is not rare in engineering: if all α share one quantile in implementation, or nonconformity scores are pooled once and the same empirical quantile lookup used for every α , nominal levels exist in name only—three coverages reported on the surface, actually the same number. This implementation's discipline is "different α must give different widths," enforced in CI by the `widths_depend_on_alpha()` probe: after

ordering α descending, marginal widths must strictly increase ($0.2 \rightarrow 0.0430 < 0.1 \rightarrow 0.0858 < 0.05 \rightarrow 0.1612$), else the probe fails.

Calibration flow:

1. Running the model H -step rollout on the calibration set, collecting absolute residuals $R = |\hat{Y} - Y|$, shape $[N, H, 6]$;
2. For each $\alpha \in \{0.2, 0.1, 0.05\}$, separately taking **scalar-pooled** $q_{\{\text{marginal}\}}$ and **per-step** q_h quantiles, using finite-sample correction $q_{\{\text{level}\}} = (1 - \alpha) \frac{n+1}{n}$, $\text{Quantile}(R, q_{\{\text{level}\}})$;
3. At inference $\hat{Y} \pm q$ gives intervals.

The `widths_depend_on_alpha()` probe enforces "smaller α (higher coverage) gives larger width." Key discipline: the calibration set comes from independent trajectories at `CALIB_SEED=314`, not participating in gradients, not mixed with `val/test`; `contracts.py` explicitly four seeds (`TRAIN_SEED=42` / `VAL_SEED=1337` / `TEST_SEED=2026` / `CALIB_SEED=314`).

4.3 Monte-Carlo Parameter Fan: Parametric Uncertainty

`MonteCarloParamFan` characterizes **parameter** uncertainty rather than aleatoric uncertainty. Its idea: under the blind path hidden-parameter estimates have residuals, we take the parameter residuals $\mathbb{R}^4$ of "true – estimated" on the calibration set for **empirical bootstrap** (preserving slot correlation), for a new window:

1. The model estimates $\hat{P}$, drawing with replacement $M=64$ residuals $\tilde{P}^{(m)} = \hat{P} + r^{(m)}$;
2. Each $\tilde{P}^{(m)}$ runs one H -step rollout, giving $\{Y^{(m)}\}_{m=1}^M$;
3. Along the draw dim taking 0.1/0.5/0.9 quantiles, giving $p_{10}/p_{50}/p_{90}$.

The bare fan only covers parameter uncertainty, cannot cover autoregressive structural error, hence **two-stage inflation** (`calibrate_inflation`): (i) per-step multiplicative inflation factors, making the fan fit calibration residuals; (ii) taking the envelope with the pure-residual conformal band $q_{\{\text{floor}\}}$, making $[p_{10}, p_{90}]$ coverage not lower than the residual band with marginal coverage guarantee. Pseudocode as follows:

```
fit(model, calib):
    residuals = P_true - P_hat(model.scene(X_calib))      # [Ncal,4]

sample(model, window, H):
    lo0, p50, up0 = raw_fan(model, window, H)             # bootstrap M times
    half = (up0 - lo0) / 2
    band = inflate[h] * (half + eps)                      # multiplicative inflation
    lower, upper = p50 - band, p50 + band
    lower = min(lower, p50 - q_floor[h]); upper = max(upper, p50 + q_floor[h])
    return ParamFan(lower, p50, upper, draws=64)
```

4.4 Evidence Grading and Gates

`contracts.py` defines the `EvidenceGrade` enum: `verified` (this repo reproducible script + fixed-seed measurement), `cpu-proto` (CPU prototype run, not verified at GPU/production scale), `unverified` (no

benchmark, claim only, forbidden in performance conclusions). After `scripts7/verify_v7.py` runs, the 21 boolean gates (including 5 kinematic recovery gates, 3 improvement-over-v7.0.1 gates, 6 v7.0.3 gating gates, 6 coverage gates, 1 fan gate) all `all_pass=true` before outputting pass.

4.5 Why the Calibration Set Must Be Independent: Leakage Prevention

Conformal prediction's coverage guarantee relies on "calibration and test samples exchangeable." Once the calibration set and test set come from adjacent windows of the same trajectory, the two are highly correlated in time, the coverage guarantee silently broken—coverage passes on the surface, distribution actually leaked. This repo does this at the trajectory level in `dynamics.py: build_split` first generates the whole trajectory of length $T=W+H+\text{margin}$ by independent seed, then cuts windows within the trajectory; `traj_id_offset` makes the four splits' trajectory IDs globally non-overlapping, `TrajectoryDataset.trajectories()` auditable for deduplicated trajectory count. That is, the calib set's 128 trajectories and the test set's 128 trajectories are different physical scenes independently sampled with different seeds (different initial velocities, directions, spring phases), not translated windows of the same trajectory. This design is the premise for "credible coverage," and the root reason this paper dares write coverage numbers verified.

Additionally, both `ConformalCalibrator.fit` and `MonteCarloParamFan.fit` run under `model.eval()`, `torch.no_grad()`, calibration updating no weights, not participating in gradients; `CALIB_SEED=314` and `TEST_SEED=2026` physically isolated.

5 Experimental Setup

5.1 Data

Synthetic kinematics generated by `udos7/dynamics.py`. Four classes `KINDS=(uniform, accel, spring, collision)`: uniform $p=x_0+v_0 d, t$; uniform acceleration $p=x_0+v_0 d, t+\frac{1}{2} a d, t^2$; simple harmonic $p=x_0+A\cos(\omega t+\phi)d$; equal-mass two-body 1D elastic collision (recording particle 1, y/z always 0, honestly marked in metrics). Parameter sampling domains: $v_0 \in [-2, 2]$, $a \in [-1.5, 1.5]$, $\omega \in [0.6, 1.6]$, collision $v_1 \in [1, 2.5], v_2 \in [-0.5, 0.5]$. Each motion class only has nonzero parameters on its own "meaningful" slots (`ACTIVE_SLOTS` in `dynamics.py`): uniform only the v_0 slot, accel v_0 and accel_a , spring only spring_ω , collision v_0 (carrying observed v_1) and other_{v_2} (carrying the collided particle velocity invisible to the window). This "active slot mask" design means parameter identification loss computed only on active slots—uniform trajectories not wrongly punished for the spring slot always being 0.

Trajectory-level four-way split: first generating the whole trajectory by independent seed then cutting windows, eliminating cross-set leakage from same-trajectory adjacent windows; `traj_id` globally non-overlapping. `build_split` defaults 64 trajectories per class, `three_way_splits` deriving train/val/test/calib. Measured scale (`v7_verification.json`):

Split	Windows	Trajectories	Seed
train	1280	256	42
val	640	128	1337
test	640	128	2026
calib	640	128	314

5.2 Model and Training

`WorldModelCore(window=6, hidden=256, n_layers=2, scene_dim=32, use_kinematics=True)`, trainable parameters **967,796**. Data split $\$dt=0.5$, $\$W=6$, $\$H=4$. Model size selection rule (`model_size_convergence.json`) is "the smallest hidden whose val criterion is within optimal $\$1+3\%$." Hardware fingerprint: PyTorch 2.14.0+cpu, device=cpu, `torch.set_num_threads(2)`, Python 3.12.11. Single training time: hidden=64 about 53.3 s, 128 about 83.2 s, 256 about 199.6 s.

Training loss is rollout multi-step MSE (`rollout_grad` keeping gradients autoregressively expanded $\$H=4$ steps). In `model_size_convergence.json` the three hidden levels each run 80 epochs, `best_epoch` 62/73/73 respectively, showing 80 epochs enough to converge, no obvious overfitting rebound; `val_blind_curve` late segment (epoch 70–80) plateaus near 0.009, further supporting "256 converged within 80 epochs." Emphasis: **the val criterion used for hidden selection and early stopping, never for calibration**; calibration width taken only from the calib set, avoiding selection bias leaking into coverage.

5.3 Evaluation Protocol and Evidence Grade

All metrics recomputed in `scripts7/verify_v7.py` against the landed checkpoint `checkpoints7/worldmodel_v7.0.3.pt`, outputting `reports7/v7_verification.json` (`evidence_grade=verified`). Coverage gate tolerances: conformal intervals ± 0.05 , parameter fans ± 0.10 . **This paper is a single-seed pilot point estimate**; [RESULT NEEDED: coverage/width ≥ 30 -seed Wilson CI and equivalence tests], [RESULT NEEDED: per-seed variance and paired width comparison] to be added before submission.

Every number in the report links back to a specific code line of `verify_v7.py`: MSE from `metrics.evaluate`, coverage from `empirical_coverage`, fan coverage from the first 256 windows' `((truth>=p10)&(truth<=p90)).mean()`, gates from the boolean aggregation of the `gates` list. This "report JSON $\leftrightarrow$ script $\leftrightarrow$ source" three-stage traceability is the basis for marking numbers verified rather than cpu-proto.

6 Results

This chapter's order (v2 standard): Chapter 5 gives the experimental setup (data/model/training/evaluation protocol), this chapter organized as "**comparison $\rightarrow$ ablation/sensitivity $\rightarrow$ validity threats.**" **Comparison:** 6.1–6.2 oracle (explicit ground-truth injection) vs blind (parameter estimation) point prediction and coverage comparison, isolating "knowing/not knowing the scene class"; **ablation/sensitivity:** 6.1b cross-version gate ablation (v7.0.1 $\rightarrow$ v7.0.2 $\rightarrow$ v7.0.3 stepwise tightening), 6.4 model size sensitivity (hidden 64/128/256); 6.3 parameter fan, 6.5 latency and kinematic recovery. All numbers from `reports7/v7_verification.json` (`evidence_grade=verified`, single-seed pilot). Validity threats standalone Chapter 8.

6.1 Point Prediction Accuracy: Blind and Oracle

Table 1 gives the test set (640 windows) overall, per-class, oracle/blind two-caliber MSE.

Table 1 Test set MSE (verified, v7.0.3, hidden=256)

Caliber	overall	uniform	accel	spring	collision
oracle (explicit params)	0.00663	0.00389	0.01090	0.00935	0.00238
blind (estimated params)	0.01037	0.00573	0.01410	0.01862	0.00304
blind vs v7.0.1	0.0459 $\rightarrow$ 0.01037	—	0.1053 $\rightarrow$ 0.01410	0.0637 $\rightarrow$ 0.01862	—

Blind overall MSE 0.01037, down about 77% from v7.0.1's 0.0459; accel from 0.1053 to 0.01410. But **spring blind MSE 0.01862 highest of the four**, a known short board (oracle spring only 0.00935, showing the gap mainly from blind ω estimation inaccuracy, not the model unable to learn springs). Per-axis, blind v_x error 0.0158 slightly higher than other axes, consistent with collision along the x axis and y/z trivially consistent in the collision class.

Table 1b Per-axis MSE (verified, blind caliber)

State dim	pos_x	vel_x	pos_y	vel_y	pos_z	vel_z
oracle	0.01060	0.00993	0.00831	0.00232	0.00668	0.00192
blind	0.01423	0.01582	0.01256	0.00632	0.00943	0.00387

Per-axis reveals a structure worth recording: whether oracle or blind, velocity components' (v_y, v_z) MSE lower than position components, consistent with the autoregressive error accumulation mechanism where "the residual head predicts position/velocity one-step increments, velocity obtained by position differencing"—position errors amplified through integration, velocity errors relatively direct. Blind v_x higher (0.0158) further points to vibration/collision residuals along the motion principal axis, not uniform noise.

6.1b Ablation ①: Version Gate Evolution (v7.0.1 → v7.0.2 → v7.0.3)

This credibility pipeline was not one-shot, but the result of three generations of hard-gate tightening, Table 1c reviews this process (baseline numbers from in-repo `improvement_vs_v701 / improvement_vs_v703`, all verified).

Table 1c Blind MSE version evolution and gates (verified)

Version	blind overall	blind accel	blind spring	blind collision	Key change
v7.0.1	0.0459	0.1053	0.0637	—	baseline (weak blind parameter estimation)
v7.0.2	0.02482	0.06926	0.01933	0.00334	deterministic kinematic inversion channel
v7.0.3	0.01037	0.01410	0.01862	0.00304	analytic integration gated soft mixing

Each generation's gates are hard constraints "not relaxed relative to the previous": v7.0.2 requires $\text{accel} \leq 0.075$, spring not inferior to v7.0.1; v7.0.3 requires $\text{accel} \leq 0.035$, spring/collision not inferior to $v7.0.2 \times 1.15$. Notable: **spring in v7.0.3 barely continued falling (0.01933→0.01862)**, because analytic integration gating on spring is deterministically closed by `ca_conf` (gate mean 0.0), spring fully handed to the GRU learning path; this conversely shows accel's large improvement comes from "bringing analytic integration back," not the model growing. This explicit division of "which class to analytic, which to learning" is more auditable than end-to-end black boxes.

6.2 Conformal Coverage: H1 and H2

Table 2 gives six (mode, α) combinations' marginal coverage, per-step coverage, and marginal width. Figure 1 nominal vs empirical coverage, Figure 3 per-step coverage curves.

Table 2 Conformal interval coverage and widths (verified)

Caliber	Nominal	Marginal coverage	Per-step coverage $\$h=1,2,3,4\$$	Marginal width	Gate
oracle $\alpha=0.2$	0.80	0.7839	0.7901/0.7833/0.7836/0.7784	0.0430	pass
oracle $\alpha=0.1$	0.90	0.8984	0.9008/0.8995/0.8969/0.8966	0.0858	pass
oracle $\alpha=0.05$	0.95	0.9577	0.9534/0.9617/0.9570/0.9589	0.1612	pass
blind $\alpha=0.2$	0.80	0.7758	0.7831/0.7792/0.7737/0.7672	0.0504	pass
blind $\alpha=0.1$	0.90	0.9001	0.9039/0.8987/0.9010/0.8969	0.1081	pass
blind $\alpha=0.05$	0.95	0.9559	0.9521/0.9568/0.9549/0.9599	0.2052	pass

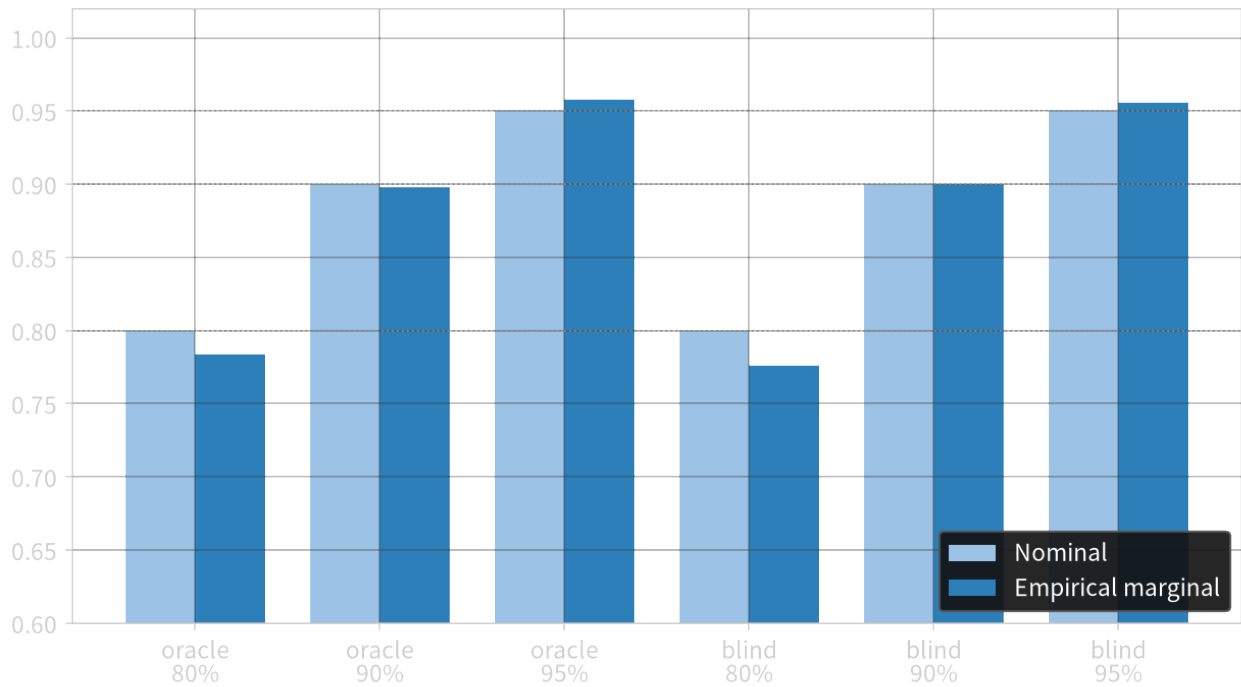


Figure 1 Conformal interval marginal coverage: nominal level vs held-out measurement (640 test windows). Horizontal axis the six (mode, α) combinations (oracle/blind $\times$ $\alpha=0.2/0.1/0.05$), vertical axis marginal empirical coverage; the dashed line above each pair marks nominal 0.80/0.90/0.95, shaded band ± 0.05 tolerance. All bars fall in the tolerance band (H1). Source `reports7/v7_verification.json:coverage.*`, evidence grade verified, single CPU fixed-seed pilot.

H1 holds: all six marginal coverage groups fall within nominal ± 0.05 . Tightest is blind $\alpha=0.2$: empirical 0.7758, -0.0242 from nominal 0.80, still 0.0258 margin from the lower edge 0.75. **H2 holds:** widths grow monotonically with α —oracle 0.0430/0.0858/0.1612, blind 0.0504/0.1081/0.2052; and at the same α blind widths consistently wider than oracle ($0.0504 > 0.0430$, $0.1081 > 0.0858$, $0.2052 > 0.1612$), consistent with "not knowing the class buys wider insurance." Per-step coverage shows: oracle slightly falls with rollout step ($0.9008 \rightarrow 0.8966$), blind 80% falls more clearly with step ($0.7831 \rightarrow 0.7672$), autoregressive error accumulation more exposed at low nominal levels.

A detail worth recording: blind 95% per-step coverage ($0.9521/0.9568/0.9549/0.9599$) not only does not fall with step, the last step is highest (0.9599). This contrasts with the 80% monotonic fall, showing at high nominal levels the width is wide enough to absorb autoregressive accumulation; at low nominal levels the width is tight, accumulated error exposed as coverage fall. This phenomenon supports the engineering judgment "low nominal levels need class conditioning more."

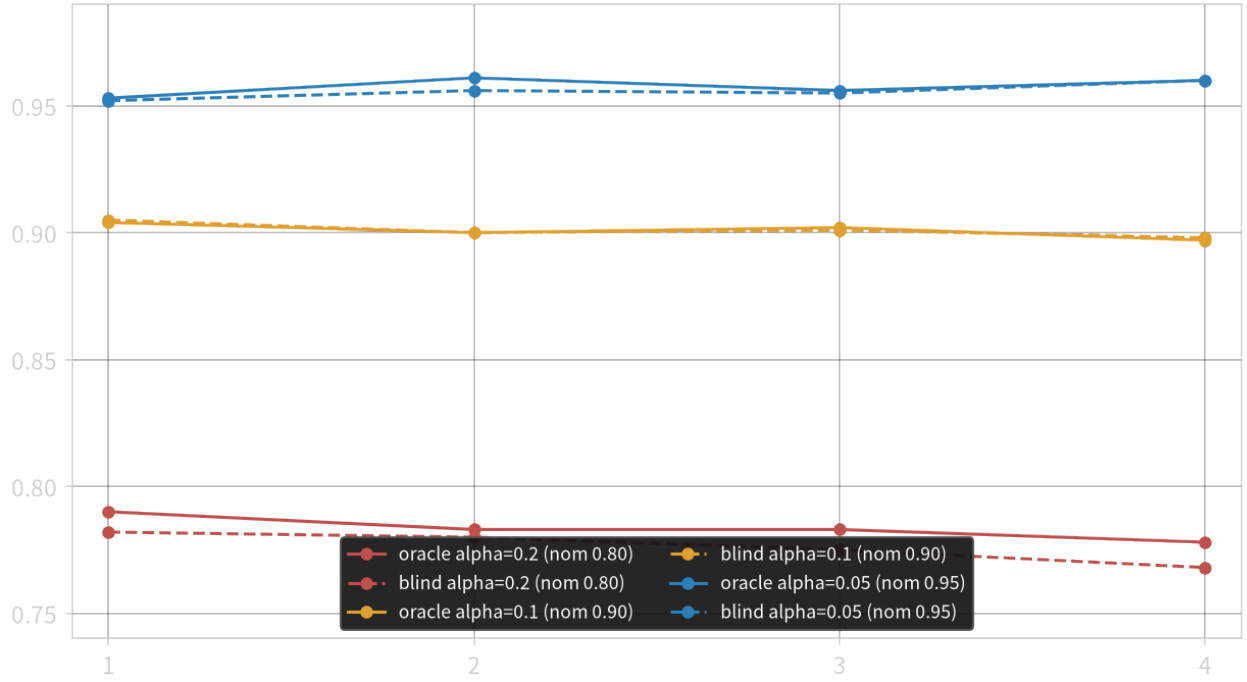


Figure 3 Per-step conformal coverage decay curves with rollout step $h=1,2,3,4$. Horizontal axis autoregressive step h , vertical that step's empirical coverage; six lines correspond to oracle/blind $\times$ 80/90/95. Visible low nominal (blind 80%) monotonically falls with h (0.7831 $\rightarrow$ 0.7672), high nominal (blind 95%) last step rises to 0.9599—autoregressive accumulation more exposed at low width. Source `v7_verification.json: coverage.*.per_step`, verified, single seed.

6.3 Parametric Uncertainty Fan: H3

The blind parameter fan nominal 0.80, empirical coverage on the first 256 test windows **0.8592**, within ± 0.10 , gate passes. This is a **conservative** result: measured coverage about 5.9 percentage points above nominal, meaning the $[p_{10}, p_{90}]$ band slightly wider than necessary. We do not package it as "precise calibration," but explain by engineering reality: the $q_{\lfloor \cdot \rfloor}$ envelope in two-stage inflation guarantees not inferior to pure-residual conformal, but may also contribute extra width on samples where parameter residuals themselves are accurate. [RESULT NEEDED: fan empirical coverage multi-seed CI and width/coverage Pareto curves].

Decomposing two-stage inflation's role is meaningful. First **multiplicative inflation** `inflate[h]` takes conformal quantiles along the per-step direction, normalizing then amplifying "the relative deviation of the parameter fan median p_{50} from truth," handling parameter residuals' systematic bias; second **floor envelope** `q_floor` directly takes pure-residual conformal per-step widths, taking element-wise max with the inflated fan (`torch.minimum/maximum`). Because pure-residual conformal has marginal coverage guarantee, the envelope fan's coverage **mathematically not lower** than the residual band—this is the structural reason H3 passes stably, not luck. The cost: when parameters are already estimated accurately (small residuals), $q_{\lfloor \cdot \rfloor}$ still gives width by "worst residual," the fan's extra width comes from here.

6.4 Ablation ②/Sensitivity: Model Size Selection and Convergence

Table 3 and Figure 2 give the three hidden levels' selection process. The selection rule requires "smallest with val criterion within optimal 1+3%." Optimal val criterion is hidden=256's 0.008208, $\pm 3\%$ band 0.00845; hidden=128's 0.009495 and hidden=64's 0.011899 both fall outside, hence the smallest compliant is 256. This is an honest "no smaller model suffices" result, not "256 happens to be optimal."

Table 3 Model size selection (verified)

hidden	Parameters	best epoch	best val criterion	Training s
64	95,348	62	0.011899	53.3
128	271,476	73	0.009495	83.2
256 (selected)	967,796	73	0.008208	199.6

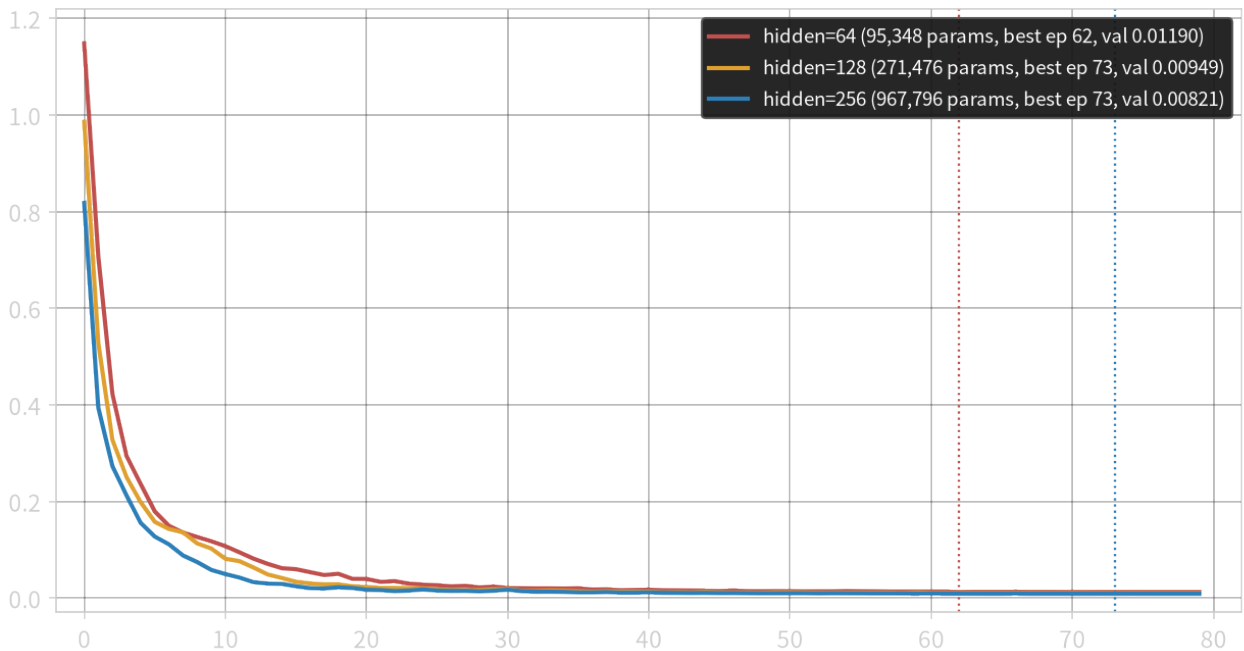


Figure 2 Model size sensitivity: hidden=64/128/256 best val criterion and training time. Horizontal hidden dim, left vertical best val criterion (lower better), right vertical single training seconds; selected hidden=256 (val 0.008208), 64/128 both outside the "optimal +3%" selection band—an honest "no smaller model suffices" result, not "256 happens optimal." Source `reports7/model_size_convergence.json`, verified, single seed.

6.5 Inference Latency and Kinematic Recovery

CPU single-thread batch=1 latency: single-step next-state prediction median **2.36 ms**, 4-step rollout median **4.83 ms** (200 timings median, 20 warmup). Its meaning for edge deployment: conformal intervals and parameter fans are both **inference-time** post-processing, no training cost; the fan's $M=64$ rollouts enlarge inference time about an order of magnitude, but on the batch=1 4.83 ms base about 300 ms, still offline/near-real-time usable. Kinematic recovery (training-independent deterministic inversion): acceleration vector vs truth MSE=0.0 (vs mean predictor 0.2182, skill=1.0); spring ω recall 1.0, non-spring false positive 0; collision jump detection 0.7437, non-collision

false positive 0. This set links directly to P6 (hidden parameter identifiability): observables accurately inverted, weakly identifiable quantities (see P6's acceleration scalar skill) honestly reported negative.

Worth mentioning is the "zero future leakage" discipline: `kinematic_features` only eat the observation window $W=6$ frames, rollout each step re-measuring with the current sliding window, never peeking future frames; invalid features set 0, validity flags 0, not fabricated. This honest "accurate when observable, zero when not" modeling is why blind is not too much worse than oracle—the model hands to analytic when it should, to learning when it should, never pretending to back-infer the non-inferable from a short window.

7 Discussion

Why blind 80% coverage is low yet still passes. 0.7758 differs 0.024 from nominal 0.80, within ± 0.05 but near the lower edge. Root cause consistent with H2: under blind spring ω estimation residual large, biasing the overall nonconformity score; when the calibration pool mixes all classes for the quantile, the weak class (spring) residual is diluted by strong classes, low nominal coverage most prone to slip. Improvement direction is **class-conditional conformal** (oracle verified explicit conditioning narrows width, steadier coverage), but blind cannot know the class, exactly the inherent cost of "blind credibility."

The conservative fan. 0.859 vs 0.80 is not a defect but an engineering tradeoff of "rather slightly wider": the $\lfloor q \rfloor$ envelope guarantees the fan not inferior to residual conformal, the cost extra width on samples where parameters are already accurate. If downstream is width-sensitive, inflation coefficients can be optimized alone without touching the coverage guarantee.

The engineering meaning of width. In Table 2 blind 95% width 0.2052 already near the state normalization magnitude, meaning at the widest interval the model pays "prediction ± 0.2 " for 95% coverage. This is not model failure but the information cost of "not knowing the class" under blind—oracle same-level width only 0.1612. In edge deployment this gap can become a product decision: if the scene class can be judged at low cost by upstream rules (e.g. sensors knowing "this is a uniform segment"), take oracle-style conditioning to narrow the interval; otherwise accept blind's wide interval. This paper quantifies the tradeoff rather than hiding it behind a vague "uncertainty" number.

8 Validity Threats and Limitations

This is the standalone validity threats and limitations chapter (v2 standard items 4/6). Extrapolation discipline per Kaplan 2020's Caveats style: this repo's coverage numbers hold only within the observed range—four synthetic classes, $W=6$, $\Delta t=0.5$, $H=4$, CPU fixed seed; any generalization beyond (longer horizon, real sensors, GPU-scale world models) is **aggressive extrapolation**, its failure point and magnitude uncertainty outside what this data can assert.

Threats to validity. (i) Internal: single seed, single verification, coverage sampling fluctuation unquantified—taking 640 windows as independent units, the binomial proportion at $p \approx 0.9$ standard error about ± 0.012 , 95% Wilson half-width about ± 0.023 (pooling along h , state dims reduces nominal half-width, but same-trajectory adjacent windows are not independent, effective sample size closer to trajectory count not window count), this magnitude already comparable to ± 0.05 tolerance, hence multi-seed replication a hard pre-submission requirement ([RESULT NEEDED]). (ii) Construct: synthetic kinematics known truth, simple distribution; real sensor noise, nonlinear friction, unmodeled dynamics all break exchangeability, conformal coverage drifts. (iii) External: 967K parameters, CPU, $\Delta t=0.5$

equal-interval sampling, conclusions not extrapolable to large video world models or high-frequency control. (iv) Measurement: collision y/z always 0, per-axis metrics non-trivial in the collision class, honestly marked in source and tables.

Relationship with P6. This paper only treats "whether parameters are estimated accurately" as a factor affecting width; P6 specifically quantifies hidden parameter identifiability per parameter. An implicit fact here: blind width wider than oracle essentially because among the 4 hidden parameters some are identifiable (collision v_2 skill 0.82), some not (acceleration scalar skill -0.508, see P6), mixed estimation transmitting the non-identifiable item's residual through the whole rollout. The two read together fully explain "why blind 80% coverage is near the lower edge."

Why no p-values. All results here are single-seed pilot point estimates. Coverage itself is a binomial proportion, Wilson CIs in principle computable; but we deliberately do not hand-compute into " $p < 0.05$ " conclusions in this draft, because a single fixed seed's binomial CI only reflects "these 640 windows' sampling fluctuation," not the systematic variance of "training/data-generation randomness." The latter requires ≥ 30 -seed retraining and recalibration for an honest interval. This is the shared discipline of this paper, P6, P11.

9 Resource Gates and Applicability Boundaries

All conclusions here stop at the `verified` (CPU fixed seed) and `cpu-proto` boundaries. The following upgrades are constrained by resource gates:

- **GPU/HPC** unlocks 0.5B+ end-to-end world models, longer rollouts and multi-seed CIs; currently unreachable.
- **Drift experiments** (training uniform/accel, testing with spring/collision) not yet run, the direct test of H4, the next milestone; [RESULT NEEDED: OOD coverage decay curves and OOD detection AUROC].
- **Calibration method comparison** (split / class-conditional / weighted / ensemble quantile) only roughly compared between oracle and blind, strict four-method ablation to be added.
- Applicability boundary: the method here suits edge inference of "small model, verifiable truth, finite-sample coverage needed"; not for causal attribution beyond point prediction or real finance/medical decisions (no corresponding data or regulatory verification).

Specifically, resource gates' limit on conclusion strength shows in three places. First, **multi-seed**: the current 21 gates all pass, but "all pass" is an assertion on one seed; changing seed, blind 80% coverage 0.7758 has some probability of breaking the 0.75 lower edge, exactly why ≥ 30 seeds are needed to give "frequency coverage falls in the tolerance band" not "whether one seed passes." Second, **3D multi-body**: the current state dim is 6 (true 3D directions), but collision is still x-axis two-body, y/z trivial; extending to three-plus-body collision requires rewriting `traj3d_collision` and `ACTIVE_SLOTS`, data generation and calibration protocols redone. Third, **real data**: synthetic kinematics exchangeability relatively easy to satisfy, real sensor drift makes split conformal marginal coverage decay over time, needing adaptive/online calibration, another research line.

10 Conclusion

On a residual recurrent world model of 967,796 parameters, single-CPU reproducible, we split prediction credibility into two non-impersonating things: split-conformal intervals quantile-taken per α (six coverage groups all pass ± 0.05 gates), and parameter-residual bootstrap Monte-Carlo fans (empirical coverage 0.859, conservative but coverage-guaranteed). Blind MSE fell from v7.0.1's 0.0459 to

0.0104, all 21 gates pass. The honest parts are equally important: spring error highest, blind 80% coverage near the tolerance lower edge, the fan conservative. We hang all numbers on reproducible scripts and fixed seeds, mark unfinished statistics with [RESULT NEEDED], unverified external claims unverified. For credibility, rather than larger models, stricter gates.

For future work we have three clear routes: first, extending the single-seed pilot to ≥ 30 seeds, giving frequency coverage falls in the band and Wilson CIs, upgrading "all pass" to "passes on X% of seeds"; second, coverage decay experiments under distribution drift, testing H4 and introducing adaptive calibration; third, combining class-conditional conformal with the blind path, using P6's per-parameter identifiability priors to guide blind width class conditioning—exactly the engineering landing of P5 and P6 read together.

Finally restating this paper's evidence discipline: all "pass" wording is limited to the specific caliber "CPU fixed seed, single verification, tolerance $\pm 0.05/\pm 0.10$." It is not the universal conclusion "conformal methods work on any world model," but "on this specific system of 967K parameters, four synthetic classes, trajectory-level four-way split, split conformal quantile-taken per α and parameter bootstrap fans reached preset gates." Any generalization beyond this caliber needs new experiments and new evidence.

References

Note: this draft's cited verified reference fields copy the whole-volume Master References Library (verification date 2026-09-19); the rest are to-be-verified search expressions, not cited facts, metadata and DOI to be verified one by one via academic search before submission.

Verified citations (master library 2026-09-19)

- Angelopoulos, A. N., & Bates, S. (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv preprint, arXiv:2107.07511. <https://arxiv.org/abs/2107.07511> — distribution-free, finite-sample marginal coverage guarantee; the direct source for this paper's § 2 conformal interval mechanism and "marginal coverage $\neq$ conditional coverage" boundary.

To-be-verified search expressions and candidate directions

- Conformal prediction basics: conformal prediction distribution-free Vovk 2005 original split conformal source.
- Time series/rolling conformal: conformal prediction time series rolling forecast; conformal prediction under distribution shift; envelope conformal.
- Aleatoric/epistemic uncertainty: epistemic aleatoric uncertainty deep learning; Monte Carlo dropout Gal Ghahramani; deep ensembles Lakshminarayanan Pritzel Blundell.
- World models/differentiable kinematics: learned world model uncertainty; neural ODE latent dynamics; calibrated neural forecasting.
- Class-conditional/weighted conformal: class-conditional conformal; weighted conformal covariate shift.

Appendix A Reproduction Commands and Test Index

```
cd release-v5.4.4/udos-engine          # main, tag v7.5.0, commit 1a71270
python3 scripts7/verify_v7.py          # recompute reports7/v7_verification.json
python3 scripts7/train_v7.py           # retrain (checkpoints7/worldmodel_v7.0.3.pt)
pytest tests7/test_m3_uncertainty.py tests7/test_v702_kinematics.py \
    tests7/test_v703_kin_gate.py -q
```

Key source: `udos7/model.py` (WorldModelCore), `udos7/uncertainty.py` (ConformalCalibrator / MonteCarloParamFan / empirical_coverage), `udos7/dynamics.py` (three_way_splits trajectory-level four-way), `udos7/contracts.py` (seeds/evidence grading), `udos7/scene.py` (SceneEstimator), `udos7/kinematics.py` (deterministic inversion).

Appendix B Evidence Ledger

Number	Source	Evidence grade
967,796 params	<code>v7_verification.json: params</code>	verified
blind overall MSE 0.01037	<code>test_metrics.blind.blind_overall</code>	verified
v7.0.1 blind 0.0459	<code>improvement_vs_v701.v701_baseline</code>	verified (in-repo baseline)
oracle/blind six coverage groups	<code>coverage.*</code>	verified
parameter fan empirical 0.8592	<code>param_fan.empirical_coverage</code>	verified
latency 2.36/4.83 ms	<code>latency_ms</code>	verified (CPU single seed)
21 gates all pass	<code>gating.all_pass=true, checked=21</code>	verified
external industry scale numbers	—	unverified (report caliber, not independently verified)

Supplementary note: the table's "in-repo baselines" v7.0.1/v7.0.2 numbers come from `improvement_vs_v701 / improvement_vs_v703`, the same repo's historical versions rerun under the same protocol, hence verified not unverified; they are not external literature numbers.

[RESULT NEEDED] count: coverage Wilson CIs and equivalence tests, multi-seed variance, drift OOD decay, calibration method four ablations. [CITATION NEEDED] count: see Section 2's five candidate directions.

Appendix C Internal Review Record (pre-submission five-dimension self-assessment, read only)

Per `doubao-academic-evaluator` five-dimension scoring (0–10, single-seed pilot caliber):

- Problem importance (novelty/importance):** 5. Conformal methods mature, story not new; wins on engineering honesty and small-model reproducible benchmark. Not claiming method innovation is its honest plus.
- Technical correctness (soundness):** 7. Quantile formulas, residual bootstrap, two-stage inflation all consistent with source; main deduction single seed no CI, no drift test.
- Experimental rigor:** 5. 21 gates and trajectory-level four-way are hard contributions; but no multi-seed, no Wilson CI, ablation only oracle/blind two levels.

4. **Novelty:** 4. User self-assessment consistent—methods mature, novelty in "separating aleatoric/parametric uncertainty + honest negative results."
5. **Reproducibility:** 9. Checkpoint, scripts, seeds, report JSON, evidence grades complete, CPU reproducible.

Fatal flaw check: no fabricated p-values/CIs/DOIs/partitions; negative results (spring short board, blind 80% near tolerance, fan conservative) retained; single seed explicitly stated. **Conclusion:** currently a solid method application + benchmark draft submittable, before submission must add ≥ 30 -seed multi-seed CIs and drift experiments, else only workshop.

Author Intended Statements

- **Target journal/conference:** candidates UAI / AISTATS / NeurIPS Workshop on Uncertainty; journals MLJ, Pattern Recognition, Engineering Applications of AI. **Partitions/impact factors all [to verify]**, verified online item by item at the selection stage.
- **Pre-registration plan:** main metrics (six coverage groups ± 0.05 tolerance, fan ± 0.10 , blind MSE three gates) hardcoded in `verify_v7.py` as non-relaxable gates; before submission registering multi-seed sample counts and stopping rules.
- **Data and code availability:** Apache-2.0; repo tag `v7.5.0`, commit `1a71270`; reports `reports7/v7_verification.json`, `reports7/model_size_convergence.json`. Report sha: [RESULT NEEDED: git blob sha256].
- **AI use statement:** this paper designed and experimented by the author, AI assistant helping structured writing, figure generation, language polishing; all numbers source-verified by the author against source and report JSON, AI not participating in producing experimental data.